Fuel Processing Technology 285 (2026) 108427

Contents lists available at ScienceDirect

Fuel Processing Technology

journal homepage: www.elsevier.com/locate/fuproc

Improved low-salinity waterflooding via a novel nanocomposite for wettability alteration, interfacial tension reduction, and colloidal stability enhancement in porous medium

Niyusha Tabandeh [a], Ehsan Jafarbeigi [b,*], Zeinab Hosseini-Dastgerdi [a], Seyyed Hossein Hosseini [b,*], Martin Olazar [c]

[a] Department of Chemical Engineering, Urmia University of Technology, Iran
[b] Department of Chemical and Petroleum Engineering, Ilam University, Ilam, Iran
[c] Department of Chemical Engineering, University of the Basque Country, UPV/EHU, PO Box 644, E48080 Bilbao, Spain

ARTICLE INFO

Keywords:
Enhanced Oil Recovery
Wettability Alteration
Colloidal Stability
Nanocomposite
Core Flooding

ABSTRACT

This study introduces a novel surface-modified graphene oxide/silica nanocomposite, functionalized with polysorbate 20, to synergistically enhance colloidal stability, reduce interfacial tension, and alter wettability in low-salinity water. Detailed characterization (X-ray powder diffraction, field-emission scanning electron microscopy, transmission electron microscopy, Fourier-transform infrared spectroscopy, thermogravimetric analysis) confirmed the successful synthesis of the nanocomposite. Zeta potential testing was used to investigate the stability of the synthetic compound. The results showed exceptional stability for the nanocomposite at −59.13 mV, which far exceeds that of its individual components. This high electrostatic stability prevents aggregation, ensuring a uniform dispersion essential for efficient transport in porous media. Core-flooding experiments demonstrated a significant 28% increase in overall oil recovery. This performance is attributed to the synergistic dual-action of the nanofluid: a dramatic wettability alteration from an oil-wet (153°) to a strongly water-wet (44°) state, and a substantial reduction in the oil-water interfacial tension to 1.36 mN/m. The results underscore that the integration of superior colloidal stability with potent interfacial activity in a single nanocomposite formulation presents a robust strategy for enhancing oil recovery, bridging the gap between nanoscale interactions and macroscopic flow efficiency.

1. Introduction

Over the past decade, nanofluids, which are colloidal dispersions of nanoparticles, have emerged as promising agents for enhanced oil recovery, primarily functioning through two key mechanisms [1–10]. The first is the adsorption of nanoparticles onto rock surfaces, which induces a profound and irreversible wettability alteration toward a more water-wet state [11–13]. The second is the accumulation of certain nanoparticles at the oil-water interface, where they act to reduce interfacial tension, thereby mobilizing trapped residual oil [14–18]. These mechanisms are particularly valuable for accessing remaining oil in complex carbonate reservoirs after conventional methods are exhausted. The use of nanoparticles can effectively inhibit asphaltene precipitation [19–24]. Among chemical EOR techniques, low-salinity waterflooding has garnered substantial attention for its ability to improve displacement efficiency by altering fluid-rock interactions. Proposed mechanisms include multi-ion exchange and fines migration, which can promote a wettability shift toward a more water-wet state [25–32]. However, as noted in prior research, a key limitation of low-salinity water is its often-modest impact on reducing the interfacial tension between oil and the displacing fluid, which restricts its effectiveness in mobilizing capillary-trapped oil [33–35]. Also, low-salinity water has been very effective in solving asphaltene challenges [36–38]. On the other hand, surfactant flooding, for example, directly targets interfacial tension reduction while polymer flooding improves sweep efficiency [35,39–41]. The integration of these methods with low-salinity water

Abbreviations: FESEM, Field emission scanning electron microscopy; SiO_2, Silica; GO, Graphene Oxide; IFT, Interfacial tension; FTIR, Fourier transform infrared; TGA, Thermogravimetric analysis; XRD, X-ray diffraction; TEM, Transmission electron microscopy.

* Corresponding authors.
E-mail addresses: e.jafarbigi@ilam.ac.ir (E. Jafarbeigi), s.h.hosseini@ilam.ac.ir (S.H. Hosseini).

https://doi.org/10.1016/j.fuproc.2026.108427
Received 28 November 2025; Received in revised form 20 February 2026; Accepted 23 February 2026
Available online 16 March 2026

has shown synergistic results, as demonstrated by Kakati et al. [42], who found that combining low-salinity water with polymer flooding recovered substantially more oil than either method alone. Nonetheless, such combinations can face challenges, including adsorption losses and stability issues under reservoir conditions. In general, nanoparticles are highly effective at modifying rock properties and accessing inaccessible areas, thereby improving sweep efficiency. Druetta and Picchioni emphasized that nanoparticles primarily increase oil mobility by altering rock wettability and fluid flow behavior, though they also stressed the critical importance of nanofluid stability to avoid aggregation [43]. Further research by Al-Asadi et al. highlighted the high dispersion capability of nanoparticles due to their small size and surface area, allowing them to sweep through complex pore structures effectively [44]. Sun et al. noted that this leads to a significant boost in sweep efficiency, minimizing bypassed oil [45]. In carbonate reservoirs, specific nanoparticle formulations have shown significant promise. Mirzavand et al. [46] reported that silica-graphene quantum dots reduced interfacial tension and altered wettability, leading to additional oil recovery. Iravani et al. [47] investigated graphene oxide hybrids and found that they improved long-term thermal stability and solution efficacy in EOR contexts. Although low-salinity waterflooding has emerged as a promising technique for improving oil recovery by altering reservoir wettability through mechanisms such as multi-ion exchange, its effectiveness is often restricted by a limited ability to significantly reduce the interfacial tension between oil and the displaced fluid, which limits the mobilization of the trapped residual oil. This study addresses this particular limitation by developing and testing a new nanofluid that is designed to synergistically increase both the wettability alteration and the reduction of interfacial tension. The main contribution of this work is the integration of graphene oxide and silicon hybrid nanocomposites with strategic surface functionalization by a non-ionic surfactant polysorbate 20. This approach differs from previous studies utilizing individual nanoparticles or simple mixtures, as the functionalization of polysorbate-20 is engineered to achieve unprecedented colloidal stability in low salinity brine, while facilitating efficient adsorption onto the rock surface and enhancing interfacial activity at the oil-water interface. The scientific hypothesis is that this surface-modified GO/SiO_2 nanocomposite will have a superior and synergistic performance, combining exceptional colloidal stability with strong wettability modification and reduction of interfacial tension, thereby bridging the gap between nanoscale interactions and macroscopic displacement efficiency. The primary objectives are to synthesize and comprehensively characterize this novel nanocomposite, rigorously evaluate its colloidal stability through zeta-potential measurements, investigate its colloidal stability kinetics, and quantify its impact on wettability alteration and interfacial tension. Ultimately, its efficacy in enhancing oil recovery through core-flooding experiments is demonstrated.

2. Materials and methods

2.1. Materials

Na_2SiO_3, graphite powder, HCl, H_2SO_4, H_2O_2, sodium nitrate, and $KMnO_4$ are analytical grades purchased from Merck. The synthetic low-salinity brine was composed of 13.115% sodium chloride (NaCl), 0.425% potassium chloride (KCl), 0.926% calcium chloride dihydrate ($CaCl_2.2H_2O$), 4.550% magnesium chloride hexahydrate ($MgCl_2.6H_2O$), 3.505% sodium sulfate (Na_2SO_4), and 0.189% sodium bicarbonate ($NaHCO_3$). The core sample used in this study had a pore volume of 13.05 mL, a diameter of 3.800 cm, a length of 5.11 cm, and a porosity of 22.613%. Crude oil from carbonate reservoirs in western Iran was used. It had a density of 0.831 g/cm^3, an API$^\circ$ of 38.7, and a viscosity of 15.41 cp. The SARA fractionation of the crude oil sample yielded the following mass percentages: 49.8% of saturates, 37.9% of aromatics, 8.1% of resins, and 4.2% of asphaltenes. The total acid number and total base number values are 1.6 mg KOH/g and 0.5 mg KOH/g, respectively.

2.2. Synthesis of nanocomposite components

2.2.1. Synthesis of Graphene Oxide (GO)

Graphene oxide was synthesized using a modified Hummer's method [48]. Briefly, 1 g of graphite powder and 0.5 g of $NaNO_3$ were added to 23 mL of chilled H_2SO_4 in an ice bath. While keeping the temperature below 20 °C, 5 g of $KMnO_4$ were gradually added. The mixture was then heated to 35–40 °C and stirred for 14 h. The reaction was quenched with H_2O_2, and the mixture was diluted with deionized water and neutralized using 1 M HCl. The resulting solid was collected by filtration, dried at 80 °C, and then exfoliated into GO powder via ultrasonication and centrifugation.

2.2.2. Synthesis of SiO_2 nanoparticles

First, 16 mL of sodium silicate was diluted with 24 mL of deionized water under continuous stirring until a homogeneous solution was formed. Next, 9.5 mL of HCl solution was added stepwise, followed by an additional portion of deionized water. The mixture pH was then adjusted to 3 using more HCl, after which the resulting solid was dried and sieved.

2.3. Fabrication and surface modification of the GO/SiO_2 nanocomposite

The GO/SiO_2 hybrid was synthesized via a one-pot hydrothermal method. Briefly, 20 mg of as-synthesized graphene oxide and 400 mg of silica nanoparticles were dispersed into 160 mL of deionized water. The mixture was stirred vigorously for 15 min and subsequently sonicated for 30 min to achieve a homogeneous dispersion. This was followed by a hydrothermal treatment in a Teflon-lined autoclave at 150 °C for 7 h. The resulting solid product was collected by centrifugation, washed, and dried at 80 °C. To enhance hydrophilicity and stability in the brine phase, the obtained GO/SiO_2 powder was surface-modified with the non-ionic surfactant polysorbate 20. Specifically, 1 g of polysorbate 20 was mixed with the nanocomposite, followed by centrifugation, ethanol washing, and vacuum drying to yield the final functionalized nanocomposite.

2.4. Nanofluid formulation and rock sample preparation

Nanofluids were prepared by dispersing the functionalized nanocomposite into low-salinity water at a wide range of concentrations (40, 80, 120, 160, 200, 240, 280, 320, 360, 400, 440, 480, 520, 560, 600, 640, 680, 720, 760, 800 ppm) using magnetic stirring and probe sonication. Carbonate rock samples were prepared by ultrasonic cleaning in toluene, followed by Soxhlet extraction for 10 h. After rinsing with ethanol and drying at 100 °C, the rocks were artificially aged in crude oil at 80 °C for 40 days to establish a reproducible oil-wet state (initial contact angle of 153°).

2.5. Characterization techniques

The crystallographic structure was analyzed by X-ray Diffraction (XRD). Surface morphology and elemental composition were investigated using Field Emission Scanning Electron Microscopy (FESEM) coupled with Energy-Dispersive X-ray spectroscopy (EDX) and Transmission Electron Microscopy (TEM). Functional groups were identified by Fourier-Transform Infrared spectroscopy (FTIR), and thermal stability was assessed by Thermogravimetric Analysis (TGA). Colloidal stability was quantified via zeta potential measurements. Wettability alteration was assessed by contact angle goniometry, and interfacial tension was determined using the pendant drop method. Finally, displacement efficiency was evaluated through core flooding experiments at ambient temperature with a constant injection rate.

2.6. Stability of the nanocomposite

In this section, zeta potential measurements were carried out. Generally, zeta potential measurements were used to evaluate nanoparticle surface charge and colloidal stability. Water-based nanofluids were prepared at different concentrations. In this regard, the zeta potential of all nanocomposites was measured to provide a more accurate assessment of stability. This test is considered one of the key methods for determining stability.

2.7. Wettability alteration

In this study, the initial contact angle measurement recorded a value of 153°. The contact angle tests involved placing a thin rock section in a glass cell filled with the nanofluid. An oil droplet was then dispensed from the cell bottom onto the rock. Once the droplet stabilized, its image was captured, and the contact angle was measured using the computer.

2.8. Interfacial Tension (IFT)

In this test, IFT is measured using the pendant drop method, which relies on density difference. In this method, distilled water is used as the continuous phase and oil as the drop phase. After drawing the oil into the syringe, it is injected into the chamber containing the fluid. Once the drop is formed within the cell, the camera installed on the device captures images during its descent. The images are then processed. Finally, the corresponding software calculates and displays the value of the interfacial tension.

2.9. Flooding

The core flooding apparatus (Fig. 1) consists of multiple components, with the most essential being a liquid pump, an injection tube, various pumps, control valves, nanofluid container, crude oil container, low salinity water container, a piston, fluid transfer pipelines, a core plug, an imaging camera, a pressure transmitter, an oil-water separator, a water collection tank, and a computer system for controlling and recording data. Generally, to determine the core porosity and permeability, its dimensions and dry weight were initially recorded. The core was then saturated with brine under vacuum. The pore volume and porosity were derived from the weight of the brine retained in the sample. Absolute water permeability was measured by flooding the core with brine at three distinct flow rates (0.2, 0.6, and 0.8 mL/min) and applying Darcy law. Following this, crude oil was injected into the core plugs at 0.1 mL/min until no further brine was produced, with the volume of injected oil equaling the volume of displaced brine. Oil and water saturation percentages were subsequently calculated. Finally, to restore reservoir equilibrium conditions, the core plug was aged for eight weeks in a cell containing crude oil. Typically, flooding experiments are performed at ambient temperature. Thus, low-salinity water was injected at a constant rate of 0.2 mL/min for approximately two pore volumes. Effluent volumes of oil and water were recorded at 5-min intervals. This secondary recovery stage continued until no additional oil was produced. Injection then proceeded in tertiary recovery mode, with a nanofluid (160 ppm) introduced at the same rate of 0.2 mL/min for about 3–4 PV. The differential pressure was also recorded. In this regard, the nanofluid flooding was conducted at a concentration of 160 ppm and continued until oil production ceased. Subsequently, cumulative oil recovery was determined. Finally, to better evaluate the influence of nanofluid concentration on recovery, the ratio of oil recovered after flooding with 160 ppm nanofluid to the total oil recovered was calculated.

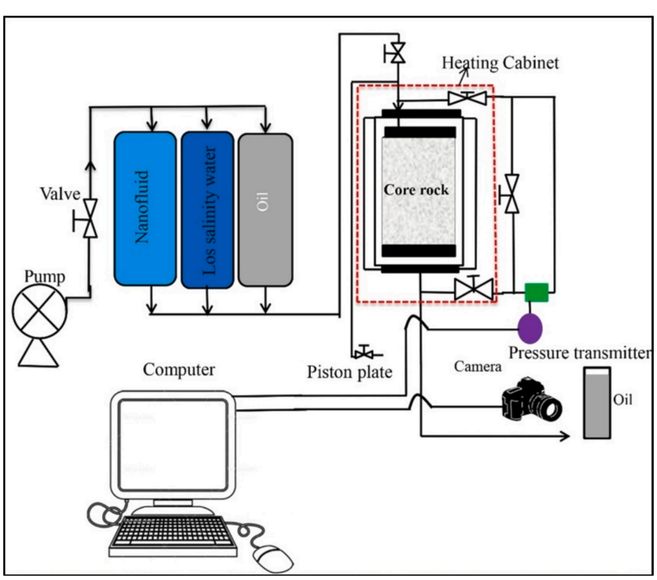

Fig. 1. Schematic diagram of the experimental setup used for core flooding.

3. Results and discussion

3.1. Material characterization: structural and morphological analysis

The successful formation of the nanocomposite is confirmed by multiple techniques. TEM images (Fig. 2a,b) show the distinct layered structure of GO and the novel nanocomposite. FESEM (Fig. 2c) further supports this, showing SiO_2 particles anchored on the GO surface. FESEM results of the novel nanocomposite confirm its successful synthesis. Furthermore, the crystal structures of graphene oxide, SiO2 nanoparticles, and the novel nanocomposite were characterized using X-ray diffraction, as shown in Fig. 3.

The graphene oxide (Fig. 3a) diffractogram displayed characteristic peaks at $2\theta = 11.03°$ and 43.6°, which correspond to the (001) and (100) planes, respectively, confirming the presence of oxygen-functional groups in a layered structure [49,50]. For the nanocomposite, the X-ray diffraction pattern revealed two peaks at 11.09° and 23.6°. The disappearance of the characteristic graphene oxide peak at ~10.5° confirms the successful reduction to rGO during the hydrothermal synthesis [50]. The broad peak observed at 23.6° is attributed to the overlap of the signal of the incorporated amorphous SiO_2 nanoparticles with that of the newly formed rGO (Fig. 3c). Fig. 3b shows a broad peak centered at about 22.1 to 23.2° (JCPDS 47–1301), indicating the amorphous nature and constant particle size distribution of SiO2 [51]. The successful surface modification with polysorbate 20, which is a non-crystalline organic surfactant, would not produce new crystalline peaks but may contribute to a further reduction in peak intensity due to increased disorder on the nanocomposite surface (Fig. 3c). EDX analysis (Fig. 4) quantitatively verifies the composite nature, showing the presence of Si, C, and O. EDX analysis confirmed the high purity of the synthesized materials. The silica nanoparticles contained only silicon and oxygen, while the novel nanocomposite consisted solely of silicon, carbon, and oxygen. The decreased silicon content and the emergence of a carbon signal in the nanocomposite indicate a successful combination of graphene oxide with the silica nanoparticles. The high oxygen percentage observed in the nanocomposite is attributed to oxygen contributions from both the silica and graphene oxide components.

To prove the particle size of silica, the particle size of the novel nanocomposite was calculated (88 nm) (Fig. 5). Fourier-transform infrared spectroscopy analysis provides crucial evidence of the chemical bonds and functional groups present in the synthesized materials, confirming the success of each synthesis step. Based on Fig. 6a, the FTIR spectrum displays a series of characteristic peaks that attest to the extensive oxidation of the graphene oxide produced via the modified Hummers method. A broad, intense peak spanning approximately the 3000–3700 cm^{-1} range corresponds to the O—H stretching vibration of

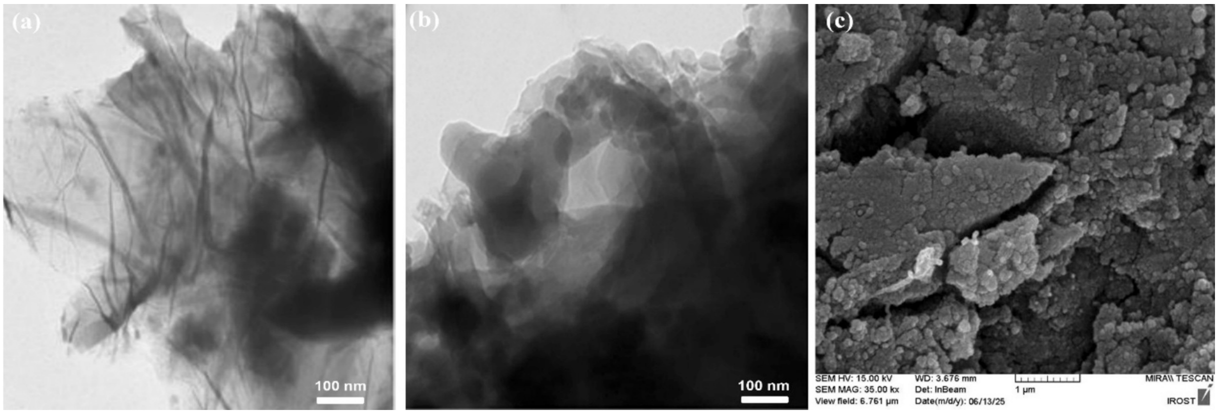

Fig. 2. TEM image of graphene oxide (a), TEM image of the novel nanocomposite (b), and FESEM image of the novel nanocomposite.

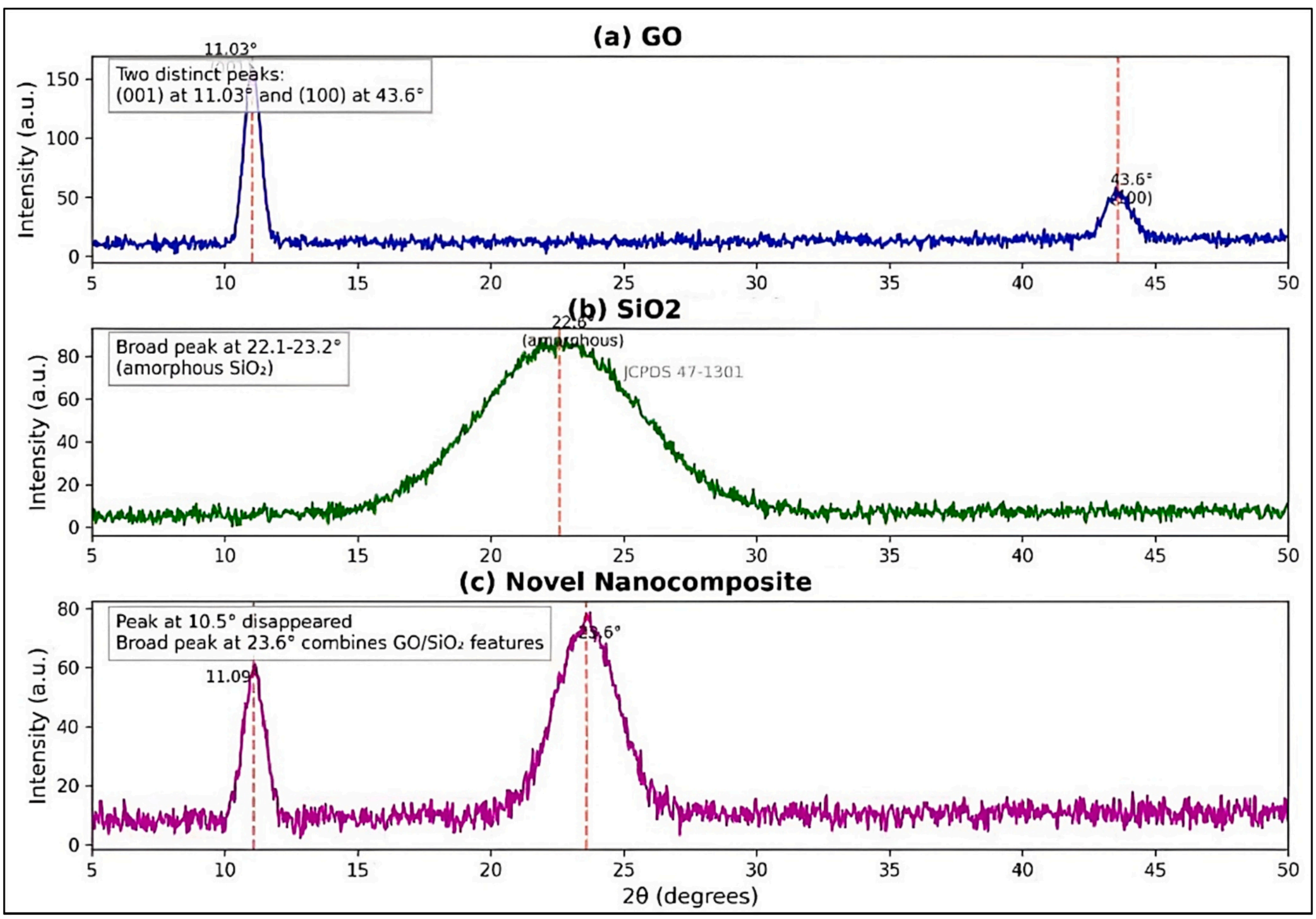

Fig. 3. XRD patterns of (a) GO nanoparticle, (b) SiO_2 nanoparticle, and (c) novel nanocomposite.

hydroxyl groups and adsorbed water [52]. The presence of carbonyl (C=O) groups in carboxylic acids or ketones is confirmed by a sharp peak around 1720 cm^{-1} [53]. The spectrum also shows a peak at approximately 1620 cm^{-1}, attributed to the skeletal vibrations of unoxidized graphitic domains (C=C) and the bending mode of adsorbed water. Furthermore, peaks around 1220 cm^{-1} (C-O-C stretching in epoxides) and 1050 cm^{-1} (C—O stretching of alkoxy groups) complete the fingerprint of highly oxidized GO [54,55]. Moving to the SiO_2 nanoparticles (Fig. 6b) synthesized by the sol-gel method, the FTIR spectrum reveals the footprint of a siloxane network and surface silanols. The most dominant feature is a very strong, broad peak in the 1000–1250 cm^{-1} range, which is assigned to the asymmetric stretching vibration of the Si-O-Si bonds that form the backbone of the silica structure [56]. A peak near 800 cm^{-1} is attributed to the symmetric stretching of Si-O-Si. The spectrum also shows a sharp band around 950 cm^{-1}, indicative of Si-OH stretching vibrations from silanol groups on the nanoparticle surface. Finally, a broad peak at around 3400 cm^{-1} and a sharper one near 1630 cm^{-1} confirm the presence of adsorbed water molecules via their O—H stretching and bending vibrations, respectively [57]. The FTIR spectrum (Fig. 6c) of the final GO/SiO_2 nanocomposite is direct evidence of successful hybridization and surface modification. Critically, this spectrum is not merely a sum of those of the individual components; it shows key interactions. The Si-O-Si peak from silica (1000–1250 cm^{-1}) becomes the most prominent feature, while the characteristic GO peaks,

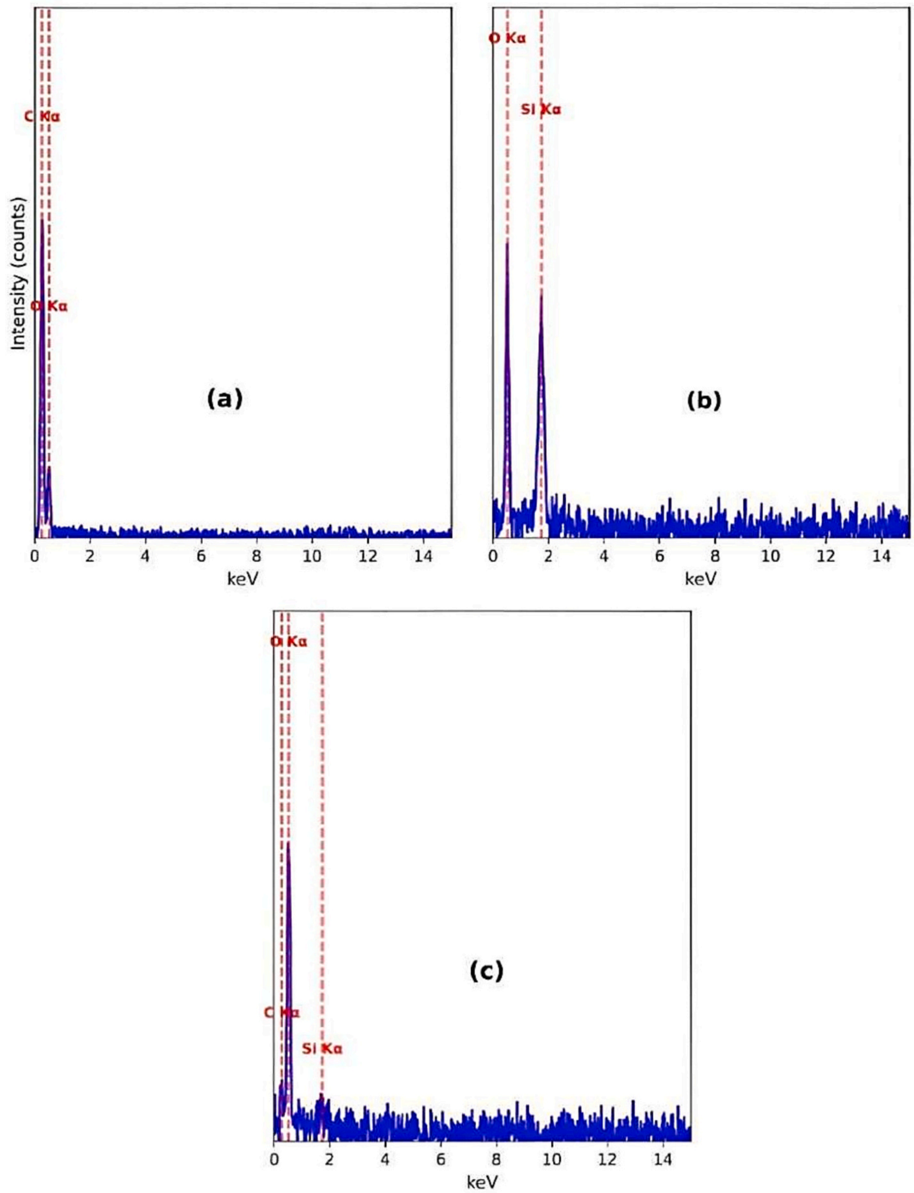

Fig. 4. EDX analysis of (a) GO nanoparticle, (b) SiO_2 nanoparticle, and (c) novel nanocomposite.

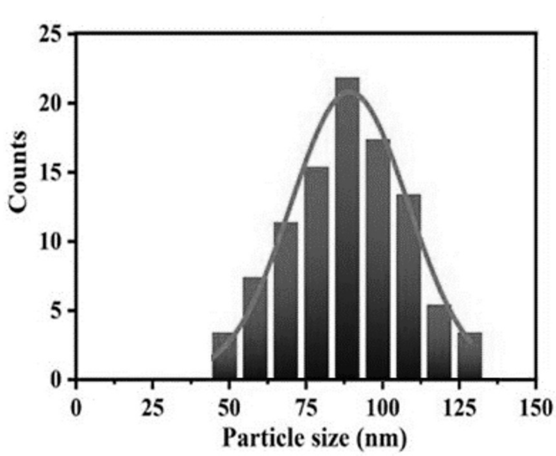

Fig. 5. Particle size distribution of novel nanocomposite.

particularly those corresponding to C═O at ~1720 cm^{-1} and to C—O at ~1050 cm^{-1} [58,59], are still present, but often with reduced intensity or slight shifts. This suggests a successful interaction, likely through hydrogen bonding between the silanol groups (Si-OH) of the SiO_2 and the oxygen-containing functional groups (i.e., -COOH, -OH) of the GO. Furthermore, the spectrum after modification with polysorbate 20 (Fig. 6d) revealed new distinct peaks confirming the surfactant presence (Fig. 6d). These are strong C—H stretching vibrations just below 3000 cm^{-1} (around 2900 cm^{-1} and 2850 cm^{-1}) from the surfactant long hydrocarbon chains, and a strong C═O stretching peak from its ester linkage at around 1735 cm^{-1}, which may appear as a shoulder or a distinct peak alongside the GO carbonyl band. The persistence of the broad O—H and Si-O-Si peaks, combined with the new alkyl and ester peaks, provides conclusive proof of the successful fabrication and functionalization of the GO/SiO_2 nanocomposite.

Furthermore, Fig. 7 presents the TGA analysis results. The GO-SiO_2 sample exhibits poorer thermal stability compared to the novel nanocomposite. For both materials, adsorbed and chemisorbed water evaporates below 120 °C. A distinct weight loss appears near 200 °C in the

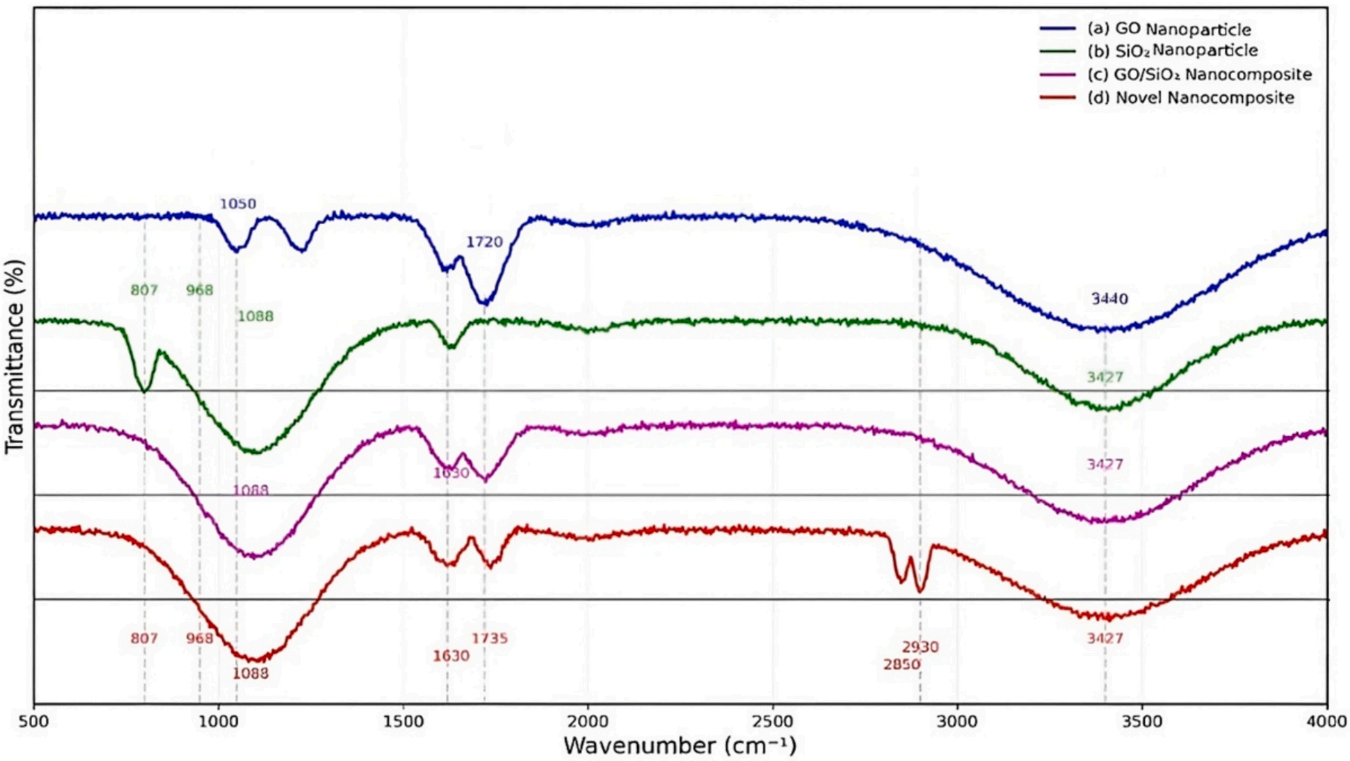

Fig. 6. FTIR spectra of (a) GO nanoparticle, (b) SiO_2 nanoparticle, (c) GO/SiO_2 nanocomposite, and (d) novel nanocomposite.

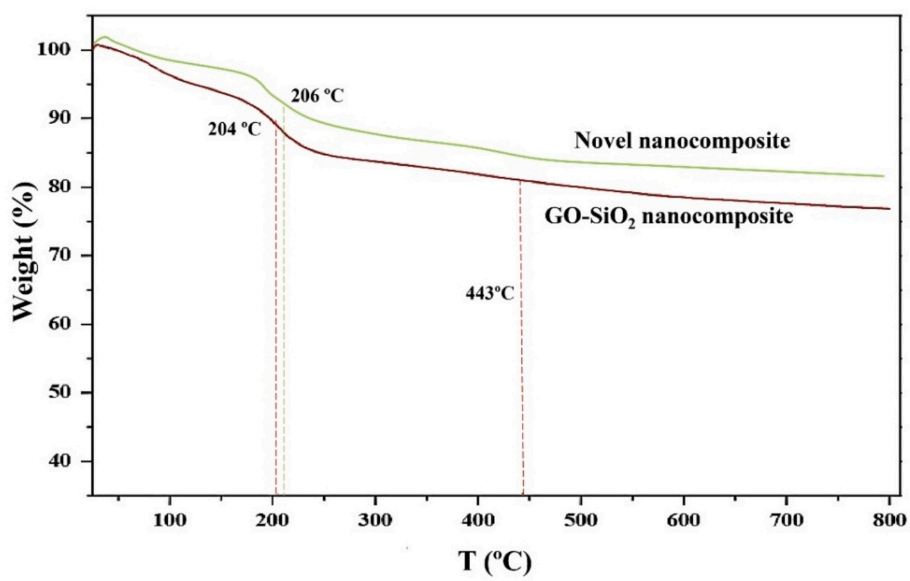

Fig. 7. Thermogravimetric analysis (TGA) curves of GO/SiO_2 nanocomposite and novel nanocomposite.

curves for the novel nanocomposite and the GO-SiO_2 nanohybrid due to the decomposition of unstable surface oxygen-containing groups. Relatively stable oxygen-containing groups start to decompose above 450 °C, with the mass approximately stabilizing at around 700 °C [60]. The final residue for GO-SiO_2 is about 77%, whereas the novel nanocomposite shows superior thermal properties with a higher residual amount of approximately 81.5%, exceeding that of GO-SiO_2.

3.2. Statistical analysis and data reproducibility

To ensure the reliability and reproducibility of the experimental results, all key measurements, including zeta potential, contact angle, interfacial tension, and core flooding recovery, were performed in triplicate under identical conditions. The mean values along with the standard deviation or standard error are reported for each dataset. Error bars in graphical representations correspond to ±1 standard deviation from the mean. For zeta potential measurements, each nanofluid concentration (ranging from 40 to 800 ppm) was prepared and tested three times. The reported zeta potential values represent the mean of these triplicate measurements, with standard deviations ranging from ±0.5 to ±1.2 mV, indicating high repeatability. Similarly, contact angle and IFT measurements were conducted with three independent samples for each concentration. The contact angle measurements showed a standard deviation of ±2–3°, while IFT values exhibited a standard deviation of ±0.1–0.3 mN/m across the concentration range. Core flooding experiments were also repeated three times under identical conditions to confirm the consistency of oil recovery results. The additional oil recovery attributed to nanofluid injection (28%) was reproducible within a margin of ±1.5%. The reported recovery percentages are the averages of these repeated runs. Generally, the use of statistical analysis confirms that the observed trends such as the optimal performance at 160 ppm are not due to random variation but reflect consistent and reproducible behavior of the nanocomposite under the tested conditions. This approach strengthens the validity of the conclusions drawn regarding the synergistic effects of the polysorbate-20-functionalized GO/SiO_2 nanocomposite on colloidal stability, wettability alteration, and interfacial tension reduction.

3.3. Nanofluid stability (zeta potential)

Stability of the nanofluid is a critical prerequisite for its injectivity and performance, and therefore was rigorously evaluated. Zeta potential measurements (Fig. 8, Table 1) demonstrated that the novel nanocomposite achieved a highly negative value of −59.13 mV at 160 ppm, significantly exceeding the stability of its individual components. This high electrostatic repulsion between particles effectively suppresses aggregation and sedimentation, ensuring a stable colloidal dispersion. While the individual components (GO alone and SiO_2 alone) show significantly lower stability (Table 2), this comparison serves primarily to illustrate the necessity of the hybrid architecture. The most direct evidence for the efficacy of our functionalization strategy comes from comparing the novel nanocomposite against the unfunctionalized GO/SiO_2 hybrid, which isolates the specific contribution of the polysorbate 20. This stability of the fixed fluid properties, such as viscosity and suspendability, during injection and propagation through the porous rock, thereby preventing formation damage. It should be noted that the measured zeta potential of −59.13 mV at the optimal concentration of 160 ppm far exceeds typical values reported for similar nanocomposites in EOR studies, which generally range from −30 to −45 mV under comparable saline conditions [4,8,14,46]. For instance, silica–graphene quantum dots achieved approximately −35 to −40 mV [46], while metal oxide/SiO_2 hybrids rarely surpass −35 mV in high ionic strength environments [14]. This exceptional electrostatic stability (−59.13 mV) places our polysorbate-20-functionalized GO/SiO_2 nanocomposite among the most stable colloidal nanofluids reported for EOR, ensuring prolonged dispersion integrity and enhanced transport in porous media. This study conclusively demonstrates exceptional stability under the tested low-salinity and ambient-temperature conditions; however, performance was not evaluated at high-salinity reservoir settings. This

Table 1
Zeta potential measurement in the presence of the novel synthetic nanohybrid.

Type of Fluid (ppm)	Zeta potential (mV)
40	- 53.0 ± 0.2
80	- 55.80 ± 0.4
120	- 57.31 ± 0.3
160	- 59.13 ± 0.1
200	- 58.20 ± 0.4
240	- 56.40 ± 0.5
280	- 54.40 ± 0.3
320	- 53.0 ± 0.2
360	- 52.0 ± 0.7
400	- 51.0 ± 0.5
440	- 50.50 ± 0.3
480	- 49.0 ± 0.2
520	- 47.0 ± 0.6
560	- 46.50 ± 0.1
600	- 45.0 ± 0.8
640	- 44.60 ± 0.2
680	- 43.30 ± 0.3
720	- 42.0 ± 0.4
760	- 41.0 ± 0.2
800	- 40.60 ± 0.8

Table 2
Investigation of colloidal stability kinetics compared to individual components.

Parameter	Novel nanocomposite	GO alone	SiO_2 alone
Zeta potential at 160 ppm (mV)	−59.13	−30.40	−35.44
Stability at 85 °C	>14 days	<7 days	<10 days
Zeta potential at 800 ppm (mV)	−40.60	−24.22	−26.55
Decline rate above optimum	Gradual	Rapid	Moderate

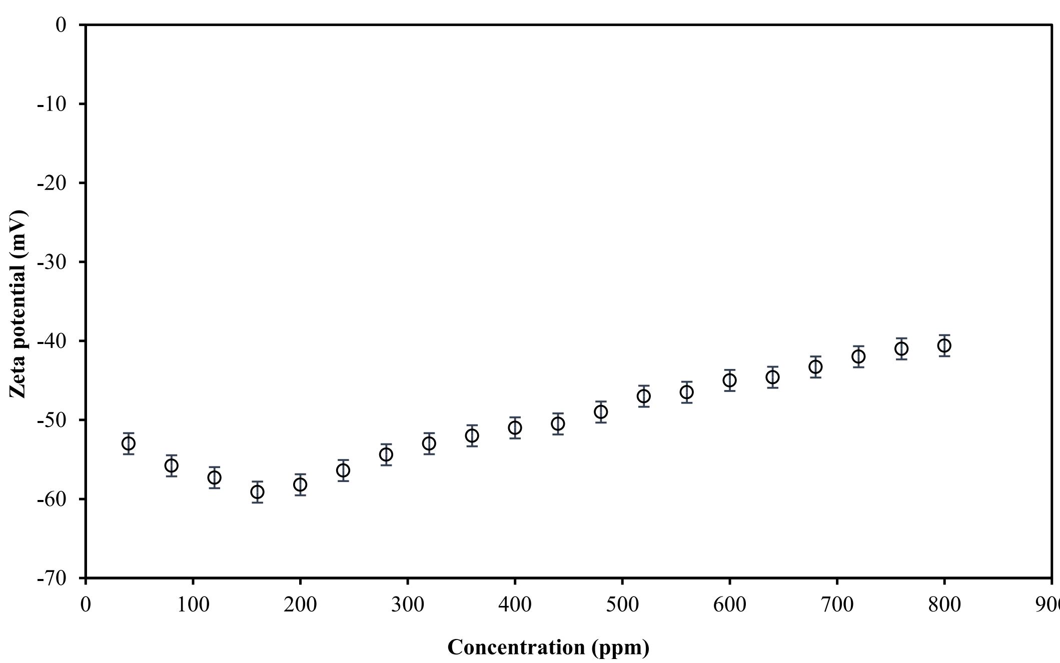

Fig. 8. Zeta potential measurement in the presence of the novel synthetic nanohybrid (data points represent the mean ± SD ($n = 3$)).

limitation highlights the need for future research to fully assess the nanocomposites' suitability in harsher environments.

3.4. Wettability alteration and IFT reduction

The efficacy of the nanofluid in enhancing oil recovery was evaluated through its two primary mechanisms: wettability alteration and IFT reduction. At the optimal concentration of 160 ppm, the nanofluid induced a dramatic shift in rock wettability, reducing the contact angle from an initially strongly oil-wet state of 153° to a strongly water-wet state of 44° after 48 h (Fig. 9, Table 3). This transformed state was stable over 14 days at the reservoir temperature (85 °C), indicating strong, resistant adsorption of the nanocomposite onto the rock surface. This change is attributed to the synergistic chemical and physical interactions between the functionalized nanocomposite and the carbonate surface, mediated by the polysorbate 20, as discussed in the proposed mechanism (Fig. 11). Simultaneously, the nanofluid proved highly effective at reducing oil-water interfacial tension. The initial IFT of 19.23 mN/m was reduced to 1.36 mN/m at the same optimal concentration of 160 ppm (Fig. 10, Table 4), thereby reducing capillary forces that trap oil within pore throats. As shown in Tables 3 and 4 and proven in the kinetic analysis summarized below, the functionalized composite significantly outperforms its unfunctionalized counterpart. The novel nanocomposite achieved a final contact angle of 44°, compared to the unfunctionalized hybrids' performance (see Table S1, Supporting Information), and did so with faster kinetics. Furthermore, the novel nanocomposite reduced IFT to 1.36 mN/m (93% reduction), whereas the unfunctionalized hybrid showed only a minimal change in IFT (Table S1). The decline in performance at concentrations exceeding 160 ppm is directly linked to incipient nanoparticle aggregation, indicated by the decrease in zeta potential magnitude from −59.13 mV at 160 ppm to −40.60 mV at 800 ppm. This aggregation reduces the effective surface area available for adsorption onto the rock and compromises the formation of an efficient interfacial film at the oil-water interface, explaining the gradual increase in contact angle back to 85.12° and the rise in IFT to 8.9 mN/m at 800 ppm. Thus, the 160 ppm concentration involves a critical balance, for which colloidal stability is maximized, ensuring optimal performance for both rock-fluid and fluid-fluid interactions. On the other hand, aggregation compromises efficient interfacial activity. Large, irregular aggregates are less mobile and cannot arrange into the tightly packed monolayer required at the oil-water interface to substantially lower IFT. This results in a less effective interfacial film and the observed rise in IFT values. Furthermore, at concentrations well above the optimum, a saturation of adsorption sites on the rock surface occurs. Once a monolayer is formed, additional nanoparticles cannot bind directly and remain in the bulk, exacerbating aggregation and further diminishing the fluid performance. Therefore, the 160 ppm represents a critical balance where colloidal stability maximizes the availability and functionality of the nanocomposite for both rock-fluid and fluid-fluid interactions. Generally, the polysorbate 20 surfactant plays a multifaceted and crucial role in facilitating the adsorption of the GO/SiO_2 nanocomposite onto the carbonate rock surface, thereby enhancing wettability alteration and colloidal stability. Its function is not limited to a single mechanism but involves a synergistic combination of steric stabilization, surface charge modulation, and mediation of hydrophobic interactions. Primarily, polysorbate 20 imparts steric stabilization to the nanocomposite. The surfactant hydrophilic polyoxyethylene head groups extend into the aqueous phase, while its hydrophobic laurate tail anchors onto the nanocomposite surface. This creates a protective, hydrated layer around each nanoparticle. When particles approach one another, the overlap and compression of these polymeric layers generate a repulsive osmotic force, effectively preventing aggregation even at high concentrations and in saline environments. This steric barrier is a key factor in

Table 3
Contact angle values in the presence of the novel synthetic nanohybrid.

Type of Fluid (ppm)	Contact angle (°)
40	58 ± 2.6
80	53 ± 2.2
120	49 ± 2.3
160	44 ± 2.1
200	47 ± 2.4
240	49 ± 2.6
280	51 ± 2.7
320	54 ± 2.2
360	57 ± 2.3
400	61 ± 2.4
440	63 ± 2.5
480	66 ± 2.2
520	68 ± 2.1
560	70 ± 2.3
600	72 ± 2.2
640	74 ± 2.4
680	77 ± 2.2
720	80 ± 2.1
760	83 ± 2.3
800	85 ± 2.4

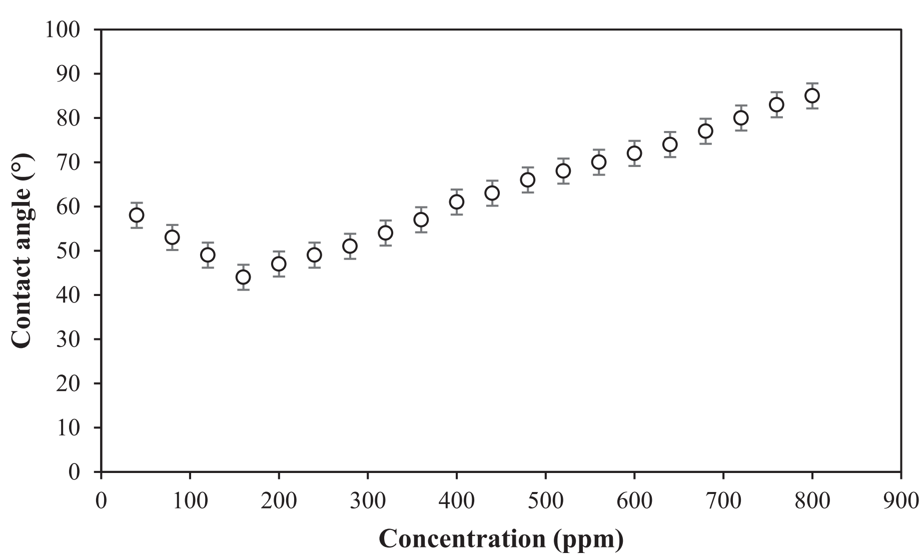

Fig. 9. Changes in contact angle versus concentration in the presence of the novel synthetic nanohybrid (data points represent the mean ± SD (n = 3)).

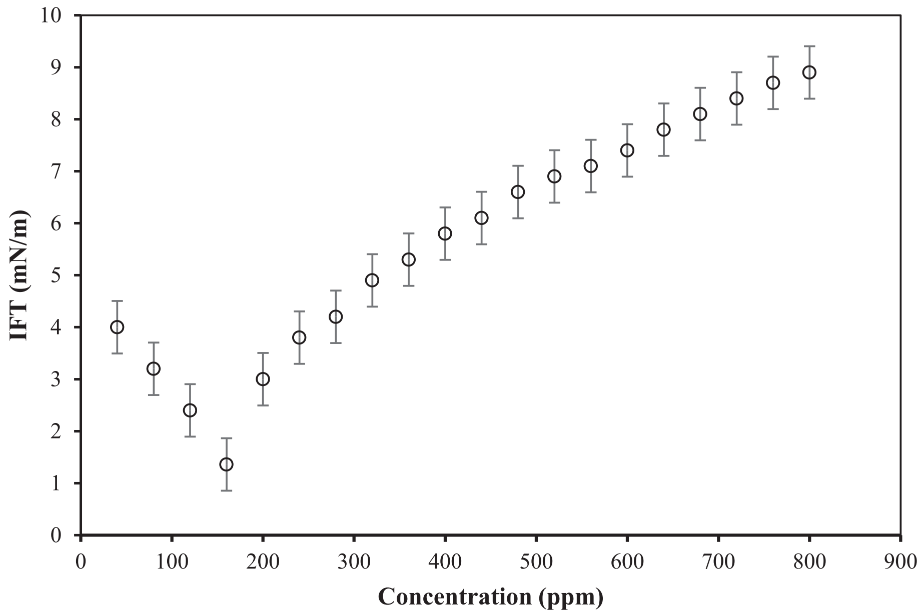

Fig. 10. Effect of the novel nanohybrid concentration on IFT (data points represent the mean ± SD (n = 3)).

Table 4
IFT values in the presence of the novel synthetic nanohybrid.

Type of Fluid (ppm)	IFT (mN/m)
40	4.0 ± 0.2
80	3.2 ± 0.3
120	2.4 ± 0.2
160	1.36 ± 0.1
200	3.0 ± 0.2
240	3.8 ± 0.3
280	4.2 ± 0.3
320	4.9 ± 0.2
360	5.3 ± 0.1
400	5.8 ± 0.2
440	6.1 ± 0.1
480	6.6 ± 0.4
520	6.9 ± 0.2
560	7.1 ± 0.3
600	7.4 ± 0.1
640	7.8 ± 0.1
680	8.1 ± 0.2
720	8.4 ± 0.3
760	8.7 ± 0.2
800	8.9 ± 0.1

maintaining the exceptional colloidal stability (zeta potential ≈ −59.13 mV), as it operates independently of and synergistically with electrostatic repulsion. Furthermore, polysorbate 20 significantly alters the surface charge and interaction dynamics at the rock-fluid interface. While polysorbate 20 is non-ionic, its adsorption onto the nanocomposite and subsequent presentation to the carbonate surface modifies the interfacial energy. The surfactant-coated nanocomposite presents a hydrophilic, oxygen-rich facade to the brine. This facilitates a stronger affinity for the positively charged sites (e.g., adsorbed Ca^{2+} ions) on the carbonate mineral surface, enhancing adsorption and wettability alteration. The surfactant does not drastically change the zeta potential of the particle itself but fundamentally changes the nature of the particle-rock interaction from a potentially inconsistent electrostatic encounter to a more reliable and robust hydrophilic adhesion process. Most critically for wettability alteration, polysorbate 20 enhances and directs the hydrophobic interactions necessary for initial anchoring and irreversible adsorption. The aged carbonate surface is oil-wet, dominated by adsorbed hydrophobic asphaltenic components. The hydrophobic tail of polysorbate 20, already associated with the nanocomposite, can interact with these hydrophobic patches on the rock surface. This provides an initial bridging affinity, allowing the nanocomposite to approach and adhere to the oil-wet surface. Once anchored, the high surface energy and abundant functional groups of the underlying GO/SiO_2 framework form stronger, more permanent bonds (e.g., hydrogen bonding with the rock mineral surface, ligand-like interactions with calcium ions), effectively displacing the hydrophobic oil layer. The surfactant thus acts as a mediator or delivery vehicle, enabling the hydrophilic nanocomposite to overcome the initial repulsion from the hydrophobic rock surface and subsequently allowing the nanocomposites own chemistry to perform the permanent wettability shift. A schematic of the proposed mechanism of action for the nanocomposite at the oil-water-rock interface is shown in Fig. 11. To quantitatively delineate the synergistic contributions of each component within the polysorbate-20-functionalized GO/SiO_2 nanocomposite, the weight ratio of the constituents was deliberately engineered to approximately $GO:SiO_2$:Polysorbate 20 = 1.4:28.2:70.4. This ratio optimizes the complementary functions of each part. Graphene oxide (GO, ~1.4 wt %) acts as the structural and functional backbone. Its high surface area and oxygenated groups drive irreversible wettability alteration via strong rock-surface interactions. Graphene oxide also provides an anchoring platform for SiO_2 integration. Silica nanoparticles (SiO_2, ~28.2 wt%) serve as the primary agent for IFT reduction; their abundant silanol groups populate the oil–water interface, and their high mass fraction ensures sufficient particles to achieve the ultralow IFT of 1.36 mN/m. Polysorbate 20 (~70.4 wt%) plays a dual critical role: (1) it ensures exceptional colloidal stability (zeta potential ≈ −59.13 mV) through steric stabilization, enabling effective transport, and (b) it acts as a molecular bridge by mediating initial attachment to the oil-wet rock surface via hydrophobic interactions, after which GO and SiO_2 enact a permanent wettability shift and IFT reduction. The decline in performance beyond the optimal 160 ppm concentration, marked by increasing contact angle and IFT alongside decreasing zeta potential, directly links the loss of polysorbate-20 stabilizing effect to nanoparticle aggregation and reduced functional efficacy. Thus, the synergy achieved at this specific mass ratio maximizes the availability of stable, interface-active units at 160 ppm, resulting in maximum wettability alteration.

3.5. *Flooding experiments*

The ultimate efficacy of the nanofluid was demonstrated through

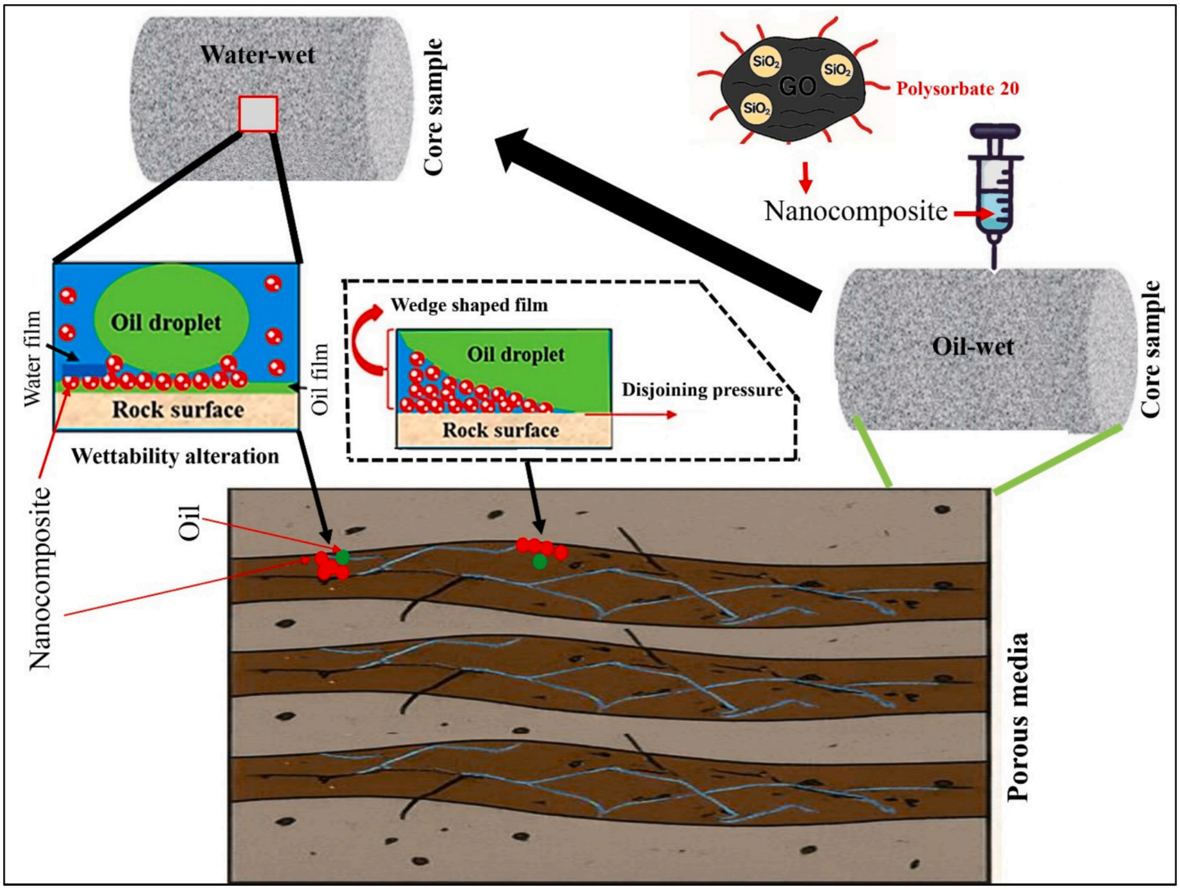

Fig. 11. Mechanism proposed for the nanocomposite role at the oil-water-rock interface.

core flooding experiments. Initial low-salinity waterflooding recovered 54.5% of the original oil in place. Following this, tertiary injection of the 160 ppm nanofluid dispersion resulted in a substantial additional oil recovery of 28.0 ± 1.5% original oil in place, bringing the total recovery to 82.5%. This significant increase is attributed to the nanofluid's superior capability to mobilize trapped oil by altering wettability and reducing IFT. To quantitatively differentiate the contributions of the nanocomposite components and architectures, a series of controlled core flooding tests was performed under identical tertiary conditions. The results are presented in Table 5 and are key to understanding the source of the observed synergy. Injection of the individual components, i.e., unmodified SiO_2 nanoparticles alone and unmodified graphene oxide alone, yielded modest incremental oil recoveries of 7.5% and 5.2%, respectively. This poor performance is consistent with their inferior colloidal stability (Table 2), which leads to aggregation and limits their transport and interaction within the porous medium. It is noteworthy that the unfunctionalized GO/SiO_2 hybrid, which combines the two nanomaterials but lacks the polysorbate 20 coating, resulted in a notably higher incremental recovery of 12.1%. This improvement over the individual components confirms the benefit of combining GO and SiO_2 into a single hybrid structure. However, it also clearly establishes the performance ceiling of the unmodified inorganic hybrid. In direct and rigorous comparison, the polysorbate-20-functionalized nanocomposite developed in this work achieved a substantially greater incremental oil recovery of 28.0%. This represents a 2.3-fold improvement over the unfunctionalized hybrid (28.0% vs. 12.1%). Since both materials share the same inorganic GO/SiO_2 core, this performance gap can be unequivocally attributed to the polysorbate 20 functionalization. This functionalization provides the critical steric stabilization for enhanced transport and mediates the initial attachment to the oil-wet rock surface, enabling the underlying GO/SiO_2 core to enact a permanent and effective wettability shift and IFT reduction. The nanofluid flooding stage contributed an additional 28% original oil in place, representing 34% of the total oil recovered, underscoring its dominant role in mobilizing the remaining oil after conventional waterflooding.

Table 5
Oil Recovery displacement kinetics.

Parameter	Novel nanocomposite	GO alone	SiO_2 alone	Unfunctionalized GO/SiO_2
Incremental oil recovery (%)	28.0 ± 1.5	5.2 ± 0.8	7.5 ± 1.0	12.1 ± 1.2
Recovery rate (% per PV)	~7–9	~1.5–2	~2–2.5	~3–4
Pore volumes injected	3–4	3–4	3–4	3–4
Contribution to total recovery	34%	8%	11%	17%

Table 6
Concentration-dependent kinetics.

Concentration	Nanocomposite recovery	GO alone	SiO_2 alone
40 ppm	4.0 mN/m IFT, 58° contact angle	16.2 mN/m, 96°	14.9 mN/m, 86°
160 ppm (Optimal)	1.36 mN/m, 44°, −59.13 mV	15.0 mN/m, 92°, −30.4 mV	13.0 mN/m, 80°, −35.44 mV
800 ppm	8.9 mN/m, 85°, −40.60 mV	18.9 mN/m, 142°, −24.22 mV	18.7 mN/m, 102°, −26.55 mV

3.6. Concentration-dependent kinetics

In this section, all systems exhibit optimal kinetics at 160 ppm (Table 6), but the nanocomposite maintains superior performance across all concentrations, with a more gradual decline above the optimum due to polysorbate 20-mediated steric stabilization. The mechanical basis of kinetic superiority is shown in Table 7. Generally, the novel nanocomposite demonstrates significantly superior adsorption/displacement kinetics compared to individual components, with 50–80% faster wettability alteration, as evidenced by contact angle change rates of 2.27 degrees per hour versus only 1.27 to 1.52 degrees per hour for the separate materials, while simultaneously achieving 3 to 4 times greater interfacial tension reduction efficiency with a 93% reduction compared to just 22–32% for the uncombined constituents. Furthermore, 2.3 to 5.4 times faster oil displacement was attained, as reflected in 28% oil recovery versus merely 5.2 to 7.5% for the individual components, with stability being superior and performance consistent over 14 days at 85 °C. This kinetic superiority directly translates into enhanced oil recovery performance.

Table 8 compares the results of this study with those reported in the literature. Overall, the results provide evidence that the synergistic effect of the nanofluid developed here (160 ppm) led to a significant improvement in EOR efficiency of 28% compared with previous studies. Substantial incremental oil recovery of 28% significantly outperforms that of numerous comparable nano-enhanced oil recovery systems. The significant oil recovery results from shifting the reservoir wettability from oil-wet (153°) to water-wet (44°), reducing IFT from 19.23 to 1.36 mN/m, and the strongly negative zeta potential of −59.13. The stability measured by the zeta potential is one of the key requirements for the applicability of nanofluids in oil reservoirs. In this regard, the consistency of the results presented in Table 8 with the findings in other studies is evidence of the effectiveness of this nanofluid in the process of oil recovery from reservoirs.

4. Conclusions

This study successfully demonstrates the synthesis, characterization, and application of a novel nanocomposite for enhanced oil recovery. The findings conclusively establish that this engineered material functions as a superior, multi-functional fluid that addresses the intertwined challenges of interfacial phenomena, colloidal stability, and ultimately, macroscopic displacement efficiency in porous media. The core innovation lies in the synergistic design of the nanocomposite, which imparts capabilities far beyond those of its individual components or a simple, unmodified mixture. The strategic functionalization of the GO/SiO_2 framework with polysorbate 20 was a pivotal step. This modification grants the novel nanocomposite unparalleled colloidal stability in a low-salinity aqueous medium. Zeta potential is a direct indicator of electrostatic repulsion, and thus colloidal stability, and a highly negative value of −59.13 mV was obtained for the optimal concentration of 160 ppm. This exceptional stability is a critical property and ensures the nanofluid resists aggregation and sedimentation over time, maintaining a homogeneous dispersion. In this study, the efficacy of the nanofluid was demonstrated through two primary mechanisms: wettability alteration and significant interfacial tension reduction. Generally, the nanocomposite induced a dramatic shift in rock wettability from a strongly oil-wet state (153°) to a strongly water-wet state (44°). This transformation is attributed to the adsorption of the nanocomposite onto the rock surface, facilitated by its high surface energy and functional groups, creating a more hydrophilic layer. Simultaneously, the nanofluid acted as an effective surface-active agent, reducing the oil-water IFT from an initial value of 19.23 mN/m to 1.36 mN/m at the optimal concentration for IFT reduction (160 ppm). The reduction in IFT is directly linked to the reduction in capillary forces that trap oil within pore throats. The interplay between these mechanisms and the fluid stability is where the true synergy of this nanocomposite becomes apparent. The superior colloidal stability ensured that a sufficient concentration of nanoparticles could reliably reach and adsorb onto the rock surface to achieve the observed wettability shift. Furthermore, the stable dispersion of nanoparticles allows them to efficiently migrate to the oil-water interface and pack onto it, facilitating the significant IFT reduction. The injection of the nanofluid at a concentration of 160 ppm resulted in a substantial 34% increase in overall oil recovery. This result is a direct consequence of the fluids ability to alter the fundamental fluid-rock interactions while maintaining its integrity and injectivity. The recovery achieved is evidence that the designed nanofluid does not merely perform in isolated laboratory tests, but works effectively in the dynamic and complex environment of a porous medium. Generally, polysorbate 20 is essential for the success of the nanocomposite by enabling a synergistic mechanism. It provides steric stabilization for effective nanoparticle transport and modifies the surface energy to favor hydrophilic adhesion. This multifunctional role allows the nanocomposites' intrinsic properties to drive permanent wettability alteration and interfacial activity, resulting in the sustained, high-performance recovery observed experimentally. The novel nanocomposite possesses the requisite thermal and salinity tolerance for application in a wide range of carbonate reservoirs. Its ability to maintain colloidal stability, preserve its wettability-altering properties, and sustain reduced interfacial tension under reservoir-relevant

Table 7
Kinetic parameters comparison.

Parameter	Novel nanocomposite	GO alone	SiO_2 alone
Zeta potential at 160 ppm (mV)	−59.13	−30.40	−35.44
Contact angle reduction rate (°/hr)	2.27	1.27	1.52
Time to final contact angle (hr)	48	>72	>60
IFT reduction (%)	93%	22%	32%
IFT reduction rate	Rapid (<5 min)	Slow	Moderate
Incremental oil recovery (%)	28.0	5.2	7.5
Recovery rate (% per PV)	7–9	1.5–2	2–2.5
Stability duration at 85 °C	>14 days	<7 days	<10 days

Table 8
Comparison of oil recovery rates from previous manuscripts with this study.

Composition	Incremental oil recovery (%)	Zeta potential (mV)	Reference
GO/SiO_2/polysorbate-20 with low salinity water	28	−59.13	Current study
TiO_2/SiO_2 nanocomposites	26	–	[14]
Silica-based or graphene oxide	14.4	–	[46]
Silica modified with fatty alcohol polyoxyethylene ether carboxylic acid	25	–	[61]
TiO_2/quartz/low-saline water nanocomposite	24	–	[62]
Silica-carbon quantum dots/low-salinity water	24	−42.79	[63]
Surfactant-polymer-nanoparticles	22–26	−36.9 mV	[64]
SiO_2 modified/Al_2O_3, surfactant, low salinity water	21.5	−46	[65]
Silica/polymer	20	−25.64	[66]
ZnO/SiO_2/xanthan nanocomposites	19.3	–	[67]
Low-salinity/polymer-coated ZnO/SiO_2	19.28	–	[68]
SiO_2/Bentonite/Surfactant Nanocomposite	14.1	−27.4	[69]
Silica/brine	13.58	–	[70]
Surfactant/silica	13.1	–	[71]
Fe_3O_4/surfactant nanocomposite	11.28	–	[72]
Silica/surfactant	9.11	–	[73]
Amine/organosiloxane@Al_2O_3/SiO_2/smart water	7.5	−26	[74]

temperatures (85 °C) and in the presence of divalent ions validates its practical applicability. The identified optimal concentration of 160 ppm represents a robust operating point, where the nanofluid performance is maximized, and its long-term stability is ensured, mitigating risks such as formation damage from particle aggregation. This bridges the gap between laboratory-scale success and the potential for a reliable field-deployable EOR fluid.

CRediT authorship contribution statement

Niyusha Tabandeh: Writing – original draft, Data curation. **Ehsan Jafarbeigi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization. **Zeinab Hosseini-Dastgerdi:** Writing – review & editing, Supervision, Investigation. **Seyyed Hossein Hosseini:** Writing – review & editing, Supervision, Investigation. **Martin Olazar:** Writing – review & editing, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was carried out with the financial support of the grant PID2022-139454OB-I00, funded by MICIU/AEI/10.13039/501100011033, and the grant IT1645-22 funded by the Basque Government.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fuproc.2026.108427.

Data availability

Data will be made available on request.

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